2022

1. Tech Talk: "Recent Developments in Image Processing"

Organized by SSN IEEE Photonics Society SB Chapter as part of IEEE Day celebrations.

- **Date:** 28 Oct 2022, 11:00 AM – 12:00 PM IST
- **Venue:** Central Seminar Hall, SSN College of Engineering, Chennai
- **Speaker:** Dr. R. Nithya, Assistant Professor, Biomedical Engineering Dept, SSN College of Engineering
- **Topics covered:** Gray level transformation, histogram processing, filtering, frequency domain filtering, hybrid filter-based image enhancement, and her research on retinal disease detection
- **Audience:** Students from multiple colleges/universities across various engineering disciplines

2. PhotonX: "Laser Tag" Program

- **Date:** 26 Oct 2023, 10:00 AM – 1:00 PM IST
- **Venue:** SSN College of Engineering, Kalavakkam, Chennai
- **Organizer:** Dr. S. Allwyn, Assistant Professor, Dept. of Biomedical Engineering

Highlights:

- Dr. Allwyn delivered a lecture on laser tag circuits, properties of light, and its device applications, followed by an engaged Q&A

- 14 student volunteers from the Photonics Society core committee — led by Chair Supraja Vaidhyanathan and Treasurer Aswathama (both 4th year BME) — conducted a hands-on workshop demonstrating a laser tag circuit using LDR sensors
- Participants built their own circuits and competed in a laser tag game
- Winners and runners-up received certificates and cash prizes
- All attendees received e-certificates

3. **Photonics Outreach Program**

- **Date:** 27 Sep 2023, 12:00 PM – 4:00 PM IST
- **Venue:** Govt. Girls Higher Secondary School, Thiruporur, Chennai
- **Organizer:** Dr. N. Venkateswaran, Student Branch Coordinator, IEEE Photonics Society SSN Chapter (Professor, Dept. of Biomedical Engineering)

Highlights:

- Dr. S. Allwyn (Assistant Professor, Biomedical Engineering) lectured 130+ 11th-grade girls on light, its properties, and device applications, sparking enthusiastic Q&A
- 14 student volunteers from the Photonics Society core committee, led by Chair Supraja Vaidhyanathan, demonstrated live experiments (IR sensor, digital microscope, spectrophotometer) in batches
- Students got hands-on time to manipulate circuits and observe changing results
- All attendees received certificates

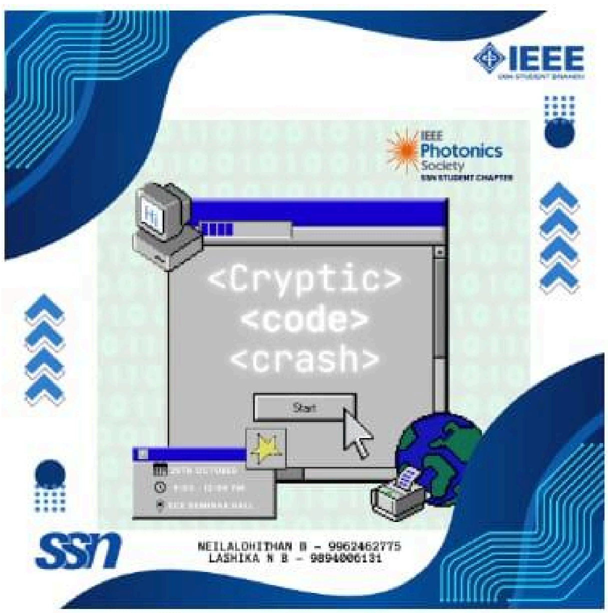
4. Cryptic Code Crash (C3) - Unveil the Mystery

- **Date:** 29 Oct 2024, 9:00 AM – 12:00 PM IST
- **Venue:** ECE Seminar Hall, SSN College of Engineering, Chennai
- **Organizer:** IEEE Photonics Society, SSN Chapter
- **Participation:** ~40 students across 15 teams

Format:

- **Round 1 – Morse Code Mania:** Multi-layered Q&A with scoring twists — +50 for correct follow-ups, -50 penalty for wrong answers, with a lifeline question worth +20. Top 6 teams advanced.
- **Round 2 – Pictionary Puzzles:** Teams sketched and decoded visual clues under time pressure, testing non-verbal communication and creativity.

Outcome: Top 2 teams (3rd-year ECE and IT students) won, receiving cash prizes for their performance across both rounds.



5. **Presentation on Photonic Devices - International Light Day**

- **Date:** 20 May 2024, 6:30 PM – 8:30 PM IST
- **Format:** Webinar
- **Venue:** SH 49A, Kalavakkam, Chennai
- **Organizer:** IEEE Photonics Student Chapter
- **Speaker:** Dr. Esther Florence S.

Highlights:

- Introduced photonics fundamentals and their everyday applications (mobile phones, projectors, medical devices)
- Explained LED principles — semiconductor physics, p-n junctions, and light generation — with healthcare applications like endoscopes and optical coherence tomography
- Covered laser sources, including optical feedback, laser oscillation, and Fabry-Perot resonators, along with medical applications in surgery
- Q&A session followed, with the speaker sharing her presentation with attendees afterward

. 6. **IEEE Photonics Society – Inauguration**

- **Date:** 28 Jul 2025, 2:00 PM – 3:30 PM IST
- **Venue:** Chemical Seminar Hall, SSN College of Engineering
- **Organized by:** IEEE Photonics Society, SSN College of Engineering
- **Coordinators:** Dr. N. Venkateswaran & Dr. S. Allwyn
- **Participants:** 2nd and 3rd year BME students (batches 2023–27 & 2024–28)

Highlights:

- Welcomed by Dr. Venkateswaran and Dr. Allwyn, who introduced the guest of honor
- Keynote by Dr. Gowri Annasamy (Assistant Professor, IIITDM Kanchipuram) on photonics applications — from biophotonics in healthcare to LiDAR in autonomous systems
- Winners of the LightScape Photography Contest were awarded certificates for their top three entries
- Badge distribution ceremony formally inducted the core committee members
- Concluded with a memento presented to the chief guest in appreciation

The event marked the official launch of the IEEE Photonics Society SSN Chapter as a platform for innovation and collaboration in photonics.


IEEE Photonics Society
GLIMPSES INTO THE WORLD OF PHOTONICS
28 JULY, 2025
2 PM - 3 PM
Seminar Hall, Chemical Department
CHIEF GUEST
Dr. Gowri Annasamy
Asst Professor @IIITDM Kanchipuram
Join us for the Inaugural Ceremony as we begin a new journey!


7. LightScape – Capture Light, Reveal Science

- **Format: Online photography contest**
- **Duration: 2 Jul 2025 – 7 Jul 2025 (7 PM submission deadline)**
- **Organized by: SSN IEEE Photonics Society, in collaboration with SSN Photography Club**

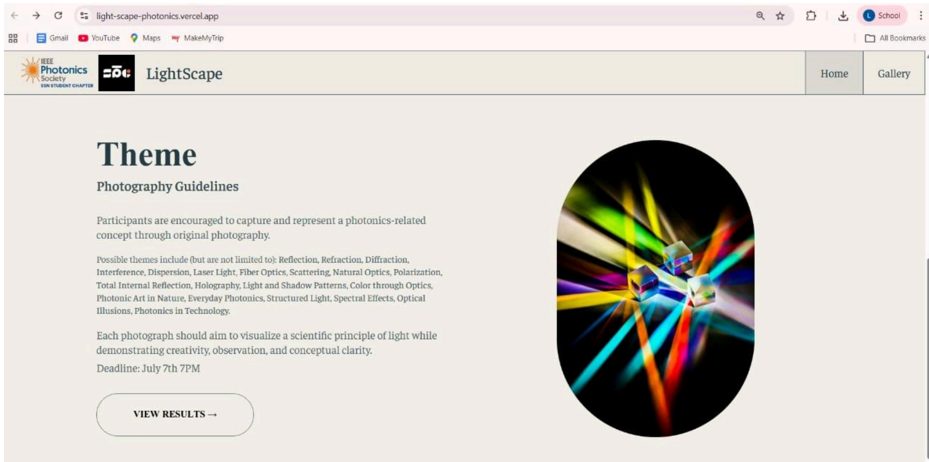
Details:

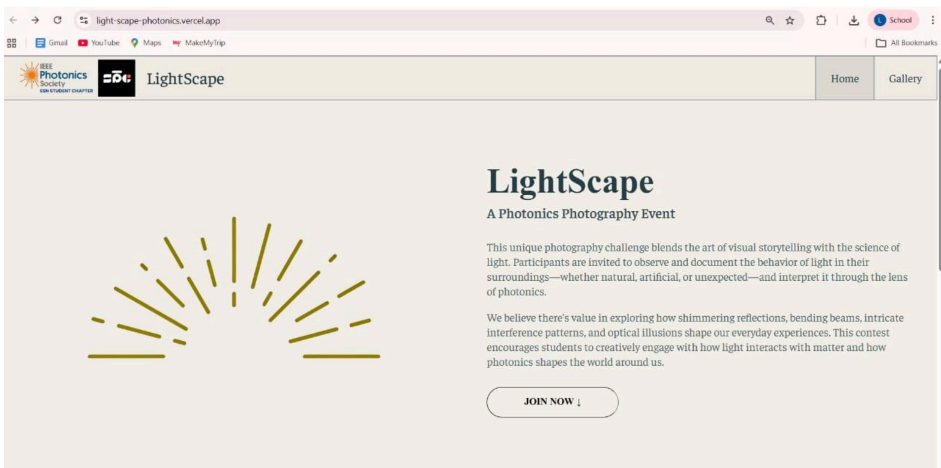
- **Theme: showcasing the role of light in science or everyday life**
- **Guidelines: one original photo per participant, basic edits allowed, AI-generated content strictly prohibited**
- **Entries submitted via a dedicated website built by the SSN Technical Team**
- **Received 22 entries evaluated on creativity, theme relevance, and concept**

Winners:

- **Best Photon Shot: Adithya Sivakumar**
- **Best Conceptual Photo: B. Raaghavan**
- **Creative Composition Award: M. Vishal Venkat**

Results were announced on the society's Instagram page. The contest encouraged interdisciplinary engagement between photonics, science, and visual art.





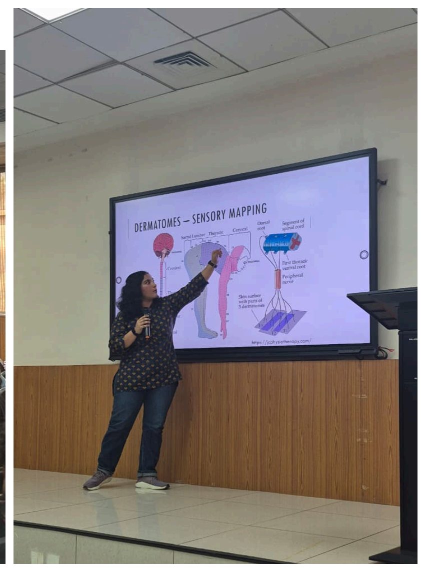
8. Mapping Neuromuscular Pathways Using Non-invasive Multimodal Techniques

- **Date:** 7 Jan 2026, 10:00 AM – 2:00 PM IST
- **Venue:** Central Seminar Hall, ECE Annexure Building, SSN College of Engineering, Kalavakkam
- **Organizers:** IEEE Photonics Society & IEEE Signal Processing Society (SPS) Student Chapters, SSN

Highlights:

- Explored challenges in neuromuscular mapping due to the system's distributed, bidirectional, multi-scale nature — no single method captures full causality, direction, and timing

- Discussed non-invasive multimodal approaches, boosted by ML/modeling, with applications in neurorehabilitation and disease phenotyping (stroke, spinal cord injury, dystonia, ALS)
- Covered pros (accessible, painless, reversible, easier ethical approval) and cons (lower spatial resolution, limited depth access, shorter recording duration) of non-invasive techniques
- Introduced brain mapping tools (EEG, fMRI, fNIRS, transcranial focused ultrasound), neuromodulation techniques (TMS, tDCS, tACS), and peripheral/spinal mapping methods (EMG, motion capture, rehabilitation robotics)
- Featured demos of exoskeletons/robots and discussion on brain-computer interfaces
- Students showed strong interest in AI/ML applications in signal interpretation and were encouraged toward research careers in neurotechnology and biomedical engineering



9. **Sensing with Light: The Evolving Landscape of Photonics Technologies**

- **Date:** 21 Jan 2026, 2:00 PM – 4:00 PM IST
- **Venue:** ECE Annexure Building, SSN College of Engineering, Kalavakkam
- **Speaker:** Dr. Srijith
- **Attendance:** ~100 students from Biomedical and ECE departments

Highlights:

- Introduced light as an electromagnetic waveform, covering amplitude, frequency, phase, and polarization in sensing/communication contexts
- Explained Fiber Bragg Grating (FBG) working principles — periodic refractive index variation reflecting the Bragg wavelength, shifted by strain and temperature
- Covered optical interrogation techniques for detecting wavelength shifts with high precision
- Showcased biomedical applications of FBG sensors:
    - Embedded sensors for continuous physiological monitoring
    - Gait cycle measurement via cuff-based sensors
    - Hand gesture analysis through wristband sensors
    - Speech interpretation via neck-region strain detection
    - Pressure/strain distribution analysis
- Sparked interest in interdisciplinary research on optical sensing and wearable biomedical technologies